

Multiple Choice (10 marks)

1. Elements with partially filled 4 f or 5 f orbitals include all of the following EXCEPT
 - (a) Cu
 - (b) Gd
 - (c) Eu
 - (d) Am
2. The strongest base in liquid ammonia is
 - (a) NH_3
 - (b) NH_2^-
 - (c) NH_4^+
 - (d) N_2H_4
3. Reduction of D-xylose with NaBH_4 yields a product that is a
 - (a) racemic mixture
 - (b) single pure enantiomer
 - (c) mixture of two diastereomers in equal amounts
 - (d) meso compound
4. Calculate the Larmor frequency for a proton in a magnetic field of 1 T.
 - (a) 23.56 GHz
 - (b) 42.58 MHz
 - (c) 74.34 kHz
 - (d) 13.93 MHz
5. The fact that the infrared absorption frequency of deuterium chloride (DCl) is shifted from that of hydrogen chloride (HCl) is due to the differences in their
 - (a) electron distribution
 - (b) dipole moment
 - (c) force constant
 - (d) reduced mass

True / False (10 marks)

Answer True or False

6. True or False: Strong oxidizing conditions are required for both paramagnetism and ferromagnetism to occur.
7. True or False: The molecular geometry of thionyl chloride, SOCl_2 , is trigonal pyramidal due to the presence of a lone pair on the sulfur atom.
8. True or False: Tetramethylsilane is not used as a reference because its protons are weakly shielded compared to other common solvents.
9. True or False: Cr_3^+ cannot be used as a paramagnetic quencher due to its electronic configuration and lack of unpaired electrons.
10. True or False: Infrared spectroscopy is a light-scattering technique commonly used in chemical analysis.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Balance the following redox reaction and discuss the stoichiometric relationship between MnO_4^- and I^- ions, providing a detailed explanation of the 6:5 ratio in your analysis.
12. Discuss the implications of the Heisenberg uncertainty principle for a quantum mechanical particle in the lowest energy level of a one-dimensional box, focusing on the limitations of knowing position and momentum.

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